

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459949.7034 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.109 +/- 0.029
Transit depth (Rp/Rs)^2	1.19 +/- 0.64 [%]
Orbital Inclination (inc)	76.91 +/- 2.98
Airmass coefficient 1 (a1)	1.152 +/- 0.021
Airmass coefficient 2 (a2)	-0.138 +/- 0.016
Scatter in the residuals of the lightcurve fit is	1.81 %
Best Comparison Star	None
Optimal Aperture	3.9
Optimal Annulus	9.59
Transit Duration (day)	0.1 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.109 $\pm$ 0.029 vs 0.1178 (deviation 0.3 σ)
Duration fit vs published	0.1 d vs 0.125 d (deviation 1.47 σ)
Mid-time O-C	-1570.9 min
Depth SNR	3.8
Residual scatter	1.81 %

Light curve

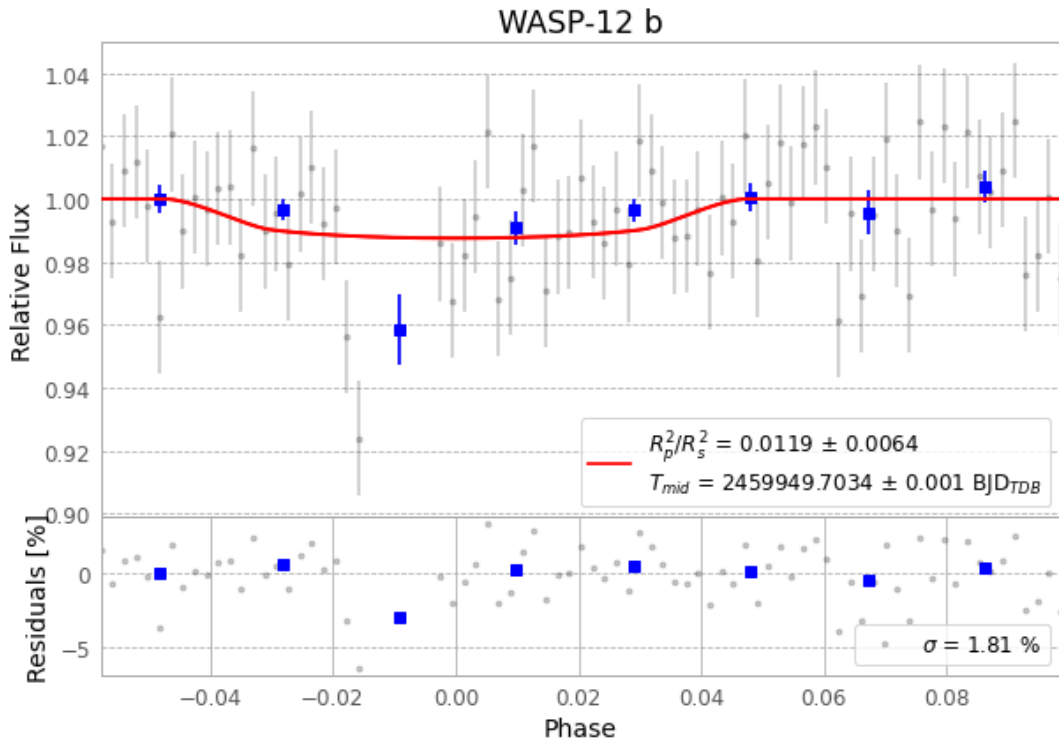


Figure 1 — FinalLightCurve_WASP-12 b_2023-01-04.png (SHA-256 9faa7e902d4c0c75...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [431, 88], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5abcf569c56ce0cb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

83 FITS frames (55 MB), WASP-12230105031216.FITS → WASP-12230105071815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-230106.log (ae855a406c99...), FinalLightCurve_WASP-12 b_2023-01-04.png (9faa7e902d4c...), FinalLightCurve_WASP-12 b_2023-01-04.csv (085bffe5644b...)

Compute resources

Wall time 85.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 05:49Z.